LESSON PLAN

A/C MATERIAL (15)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	01
Total No. of Lesson Plans	16	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Classification of aircraft, arrangements of major components and their functions
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Classification of aircraft, arrangements of major components and their functions
5. Specific Objectives	15.1.1.1 Trainee will be able to explain aircraft structure, various structural elements and their details.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of component diagrams; Safety: No physical handling of aircraft parts; Supervision: Instructor monitors diagram interpretation
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What are the primary categories of aircraft based on propulsion?
(b) Name the four major structural components of an aircraft.
(c) Define the function of the fuselage in an aircraft structure.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Classification by propulsion type • Classification by wing configuration • Classification by landing gear • Classification by mission role • Classification by size and range	L & Ds	PPT, IB	15 min
(b)	• Fuselage structure and elements • Wing structure and elements • Empennage components and functions • Landing gear types and functions • Powerplant arrangement and systems	L & Ds	PPT, PDF	20 min
(c)	• Integration of major components • Structural load paths explanation • Component interaction during flight phases • Review of specific structural elements • Summary of classification criteria	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how the wing structure supports the aircraft during flight.
(b) Differentiate between fixed and retractable landing gear arrangements.
(c) Describe the function of the horizontal stabilizer in aircraft control.

12. Home Assignment and Follow Up	Prepare a brief note on the design requirements for the next lesson (LP 02) focusing on how classification influences manufacturing processes.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Steps in aircraft design, manufacturing process and aircraft design requirements	Refer to the master syllabus reference list	LP 02

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Instructor, Flight Lieutenant, P No. 12345	Senior Instructor, Commander, P No. 67890
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min